NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Computer Science & Engineering
MID SEMESTER EXAMINATION, Oct. 2024

Semester- 05
Code: CS54117

B. Tech. (UG)
Course Name: Internet Of Things

Maximum Time: 2 hours **Max. Marks: 30**

Instruction:

1. *Attempt All questions.*
2. *You may answer the questions in any order. However, all parts of the same question must be answered together.*
3. *Assume any suitable data, if necessary. Clearly state any reasonable assumption you make.*
4. *The Marks, CO (Course Outcome), and BL (Bloom's Level) related to questions are on the right-hand margin.*

	Marks	CO	BL
1. A farmhouse needs to be monitored and controlled using IoT. The site has a small garden, a house with ACs, washing machines and fire fighting systems; a parking area for multiple cars and surveillance systems. All the data will be stored on the cloud in the Internet. Design IoT network for the site. Draw the design showing the components like IoT nodes, IoT LAN, and Gateways.	10	CO 1	A
2. In an IoT framework, nodes collect data like temperature, humidity, light brightness, temperature in the AC rooms, air pressure in some machines and ingredient quantity in coffee machines. There are different kinds of users interested in different things like agriculture, office maintenance and coffee supply in the office. Identify a suitable communication model for the design of the framework. Justify your answer. Use diagrams as needed to explain your answer.	3+7	CO2	
3. a.) Define Sensor and Transducer. Classify sensors based on their input and output in different classes? Explain Hysteresis Error and Quantization Error. b.) Define the Internet of Things and write its characteristics.	6+4	CO 1, CO 2	A

****End of Questions****

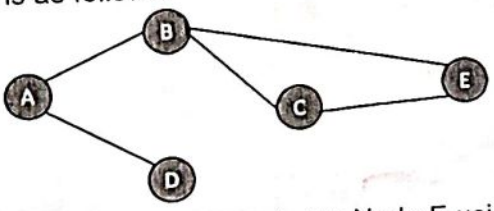
NATIONAL INSTITUTE OF TECHNOLOGY PATNA
Department of Computer Science & Engineering
End SEMESTER EXAMINATION, Dec. 2024

B. Tech. (UG) **Semester- 05**
Course Name: Internet Of Things **Code: CS54117**

Maximum Time: 3 hours **Max. Marks: 60**

Instruction:

1. *Attempt All questions.*
2. *You may answer the questions in any order. However, all parts of the same question must be answered together.*
3. *Assume any suitable data, if necessary. Clearly state any reasonable assumption you make.*
4. *The Marks, CO (Course Outcome), and BL (Bloom's Level) related to questions are on the right-hand margin.*

	Marks	CO	BL
1. You are going to design an IoT network which demands Low power consumption, Low bandwidth, Scalability, Reliability, Mobility and long sleep times. The network devices may also communicate using the Internet.		CO5 CO2	A
a.) Suggest a proper addressing technique you will choose for the network.	2		
b.) Justify your choice.	2		
c.) If the network uses 6LoWPAN; how will the trade-off between constraints of low-power devices with the requirements of IPv6 be handled?	4		
d.) "header compression critical in 6LoWPAN", justify the statement and also suggest some solution.	4		
2. Consider a Zigbee Mesh network with five nodes: A, B, C, D, and E. The network topology is as follows:  Suppose Node A needs to send a packet to Node E using AODV.		CO4	A, E
a.) Explain the RREQ messages propagation for this communication. How many RREQ Messages are required for the whole communication.	3		
b.) Show the route table entries and its updates at every node.	3		
c.) Show the RREP message propagation from E to A.	3		
d.) If each RREQ or RREP message transmission consumes **1 unit of energy** per hop, then calculate the total energy consumed during the RREQ and RREP phases.	3		

3.	An IOT application requires a short range communication which is supported by most smartphones, tablets, and computers. The application generates both streaming and periodic data. **a.)** Identify which communication technology is most suitable for such a scenario and why? **b.)** Which topology will be best suited in this scenario. **c.)** Draw the protocol stack used for your chosen communication technology. **d.)** Explain the steps involved from device discovery, connection creation to data transfer with respect to each layer of the protocol stack.	3 2 3 4	CO4 CO2	A, E
4	A smart logistics system is being designed to manage inventory, monitor shipments, and enable secure access to storage units. The system must leverage **Z-Wave**, **RFID**, and **NFC** to optimize various operations. Here are the key requirements: 1. **Inventory Tracking**: Products in a warehouse are tagged for identification and stock updates. 2. **Shipment Monitoring**: Packages need to communicate their environmental conditions (e.g., temperature, humidity) during transit. 3. **Secure Access**: Workers need a reliable and secure method to unlock storage units for product handling. **a.)** Assign Z-Wave, RFID, and NFC to specific tasks (inventory tracking, shipment monitoring, and secure access). Justify your choice of technology for each task. **b.)** Design a flowchart showing how Z-Wave, RFID, and NFC interact within the logistics system to perform their respective roles efficiently. **c.)** Analyze the advantages and limitations of using each technology for the assigned task. **d.)** Explain the difference between RFID and NFC.	3 3 3 3	CO5, CO1	U, C
5	An IoT-enabled autonomous vehicle integrates key functionalities such as temperature control, automatic wipers, and automatic lights. The system leverages sensors, cloud communication, and real-time data processing to enhance the vehicle's performance and ensure a seamless driving experience. **a.)** Identify suitable sensors for the following functionalities and justify your choices: • Monitoring cabin temperature. • Detecting rain, moisture, or road debris on the windshield. • Measuring ambient light levels to control automatic lights. **b.)** Recommend an appropriate application-layer protocol for transmitting real-time sensor data to a cloud platform. Explain your choice based on its suitability for the autonomous vehicle scenario. **c.)** Illustrate the IoT protocol stack used in this system. For each layer of the stack, explain its role and the relevant components or protocols that support the system's functionalities.	3 3 6	CO5, CO4	U, R, A

****End of Questions****